Note

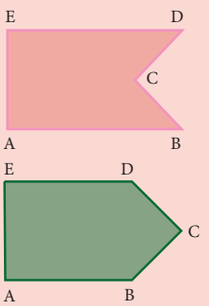
We can split any quadrilateral into two triangles by drawing a diagonal.



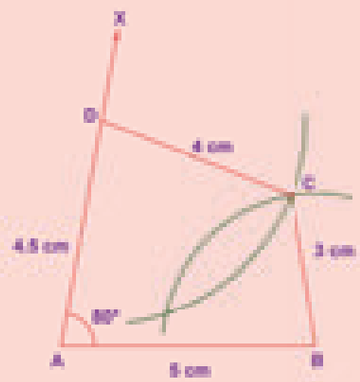
In the above figures, a quadrilateral is split in two ways by its diagonals. So, if a diagonal is given, first draw the lower triangle with two sides and one diagonal. Then, draw the upper triangle with other two measures.

DO YOU KNOW?

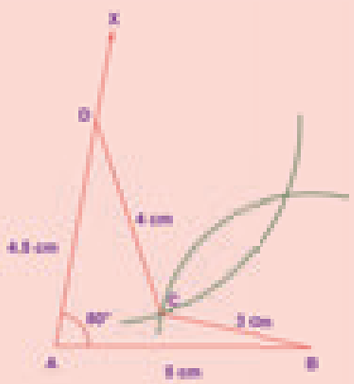
(i)
- A polygon in which atleast one interior angle is more than 180°, is called a concave polygon. In the given polygon, interior angle at C is more than 180°.
- A polygon in which each interior angle is less than 180°, is called a convex polygon. In the given polygon, all interior angles are less than 180°.



(ii) Look at the following quadrilaterals.



Convex quadrilateral



Concave quadrilateral

Though, we can construct a quadrilateral in two ways as shown above, we do not take into account the concave quadrilaterals in this chapter. Hence, the construction of only convex quadrilaterals are treated here.

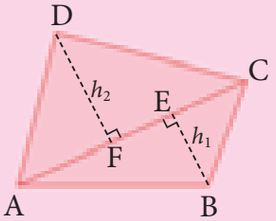
Note

Consider the given Quadrilateral ABCD. In which AC is a diagonal (d), BE (h_1) and DF (h_2) are the perpendiculars drawn from the vertices B and D on diagonal AC.

Area of quadrilateral ABCD = Area of ΔABC + Area of ΔACD

$$= \frac{1}{2} \times AC \times BE + \frac{1}{2} \times AC \times FD$$

$$= \frac{1}{2} \times d \times h_1 + \frac{1}{2} \times d \times h_2 = \frac{1}{2} \times d \times (h_1 + h_2) \text{ sq.units.}$$



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